ChatGPT

Quantum Annealing for Optimization: Key Papers

Logistics and Transportation

- **Supply chain logistics with quantum and classical annealing algorithms** – *Weinberg et al. (2023)*: A multi-truck vehicle routing problem (VRP) was tackled by iteratively solving smaller single-vehicle routing QUBOs (~2500 binary variables) using simulated annealing and the D-Wave Hybrid solver. Each step assigns one truck's route; the resulting routes were fed into a classical supply-chain simulator. The hybrid approach yielded excellent end-to-end performance on a real corporate logistics problem [1]. (Sci. Rep.) link.
- **Quantum annealing-based route optimization for automated guided vehicles (AGVs) in warehouses** – *Śmierzchalski et al. (2024)*: The scheduling of AGVs in a factory (subject to space/deadlock constraints) was mapped to a quadratic optimization problem and solved with D-Wave's hybrid solvers (classical heuristics + QPU). This hybrid method's solution quality was comparable to classical ILP solvers. In a scenario with up to 21 AGVs (significant deadlock potential), the hybrid solver found schedules in seconds [2] [3]. (Sci. Rep.) link.
- **Quantum annealing for the two-level facility location problem** – *Ciacco et al. (2026)*: Addresses a logistics NP-hard problem of locating facilities on two levels. The authors formulate it as a QUBO and solve it on a D-Wave quantum annealer. To handle large instances, they propose a preprocessing method to shrink the network before annealing. Computational experiments show that the preprocessing enables the QA approach to scale better, validating the efficacy of QA on this supply-chain problem [4]. (Future Gen. Comput. Syst.) link.
- **Quantum annealing approaches to the shipment rerouting problem** – *Li et al. (2025)*: Studies the *Shipment Rerouting Problem* (SRP) where goods must be delivered between hubs by trucks. SRP generalizes sequencing/packing NP-hard problems. The paper presents novel MILP and QUBO formulations for SRP and implements both classical algorithms and a D-Wave quantum annealing solver. Experiments show that, under various scenarios (capacitated trucks, network infrastructure), the QA-based solver finds near-optimal or optimal solutions much faster than classical solvers [5]. (ArXiv, Jan 2025) link.
- **Minimum loss problems in electrical distribution networks** – *Silva et al. (2023)*: Formulates the problem of reconfiguring power distribution networks to minimize energy losses as a QUBO. Using a D-Wave hybrid solver on the IEEE 33-node test network, they compare solutions to classical methods. The QA approach achieved competitive (near-optimal) loss reductions and showed promise for faster solution times (important for real-time reconfiguration) [6]. The authors conclude that QA may soon outperform classical solvers on such grid optimization tasks as hardware improves. (Sci. Rep. 13:10777, 2023) link.

Scheduling Problems

- **Solving the resource-constrained project scheduling problem with quantum annealing** – *Pérez Armas et al. (2024)*: Tackles the NP-hard RCPSP. Twelve MILP formulations were surveyed and the most qubit-efficient was converted to QUBO. The QUBO was solved on a D-Wave Advantage 6.3 QPU and compared to classical solvers. Results on benchmark instances (20–80 tasks) show the QA approach performs well on small/medium cases and provides insights into annealing schedules and Q-score benchmarking [7]. This is the first reported application of QA to RCPSP. (Nat. Sci. Rep.) link.

- **Evaluating the job shop scheduling problem on a D-Wave quantum annealer** – *Carugno et al. (2022)*: Presents a full pipeline for JSSP (jobs–machines scheduling) on a D-Wave QPU. The JSSP is formulated as a QUBO, then embedded and solved on D-Wave QPUs. They analyze solution quality and embedding strategies on instances with up to 10 jobs. The study is among the first to systematically test QA on JSSP and reports that, while challenging, the QPU could find feasible schedules and provides groundwork for future hybrid approaches [8]. (Sci. Rep.) link.
- **A hybrid quantum annealing heuristic for job shop scheduling** – *Kurowski et al. (2020)*: Experiments with JSSP benchmarks using D-Wave 2000Q (2048-qubit QPU). The authors decompose each JSSP instance into smaller QUBOs to fit the hardware, tuning parameters of the limited qubit graph. This proof-of-concept shows how current QA hardware can be used via a heuristic: by iteratively solving subproblems and combining results. The paper discusses how future D-Wave improvements could enhance performance [9]. (Electronics) open access link.
- **Quantum annealing implementation of job-shop scheduling** – *Venturelli et al. (2016)*: An early study mapping the *Job-Shop Scheduling Problem (JSP)* to a time-indexed QUBO and solving it on the D-Wave Vesuvius (512 qubits). The authors apply variable pruning and binary search heuristics to schedule up to 6 jobs on 6 machines. They compare QA results to optimal classical solvers, demonstrating that QA can find valid schedules on small benchmarks. This work details preprocessing and embedding strategies for QA-based JSP solutions [10]. (COPLAS workshop 2016; ArXiv) link.
- **Hybrid quantum-classical computation for AGV scheduling** – *Śmierzchalski et al. (2024)* (see Logistics above) – addresses scheduling in manufacturing (AGVs) and is also a scheduling use-case.

Finance and Portfolio Optimization

- **Portfolio optimization using the D-Wave quantum annealer** – *Phillipson & Bhatia (2020)*: Considers a Markowitz-style portfolio (60 stocks from Nikkei225 and S&P500). They formulate the mean-variance problem as QUBO and solve with D-Wave annealers and hybrid solvers. For the tested subset sizes (20–60 assets), the D-Wave solutions nearly match those of commercial solvers in return/risk trade-off. This study shows that current QA hardware can handle modest portfolio sizes with results similar to classical optimization [11]. (ArXiv, 2020) link.
- **End-to-end portfolio optimization with quantum annealing** – *Morapakula et al. (2025)*: Implements a full pipeline for constructing and rebalancing an investment portfolio. A continuous mean-variance model is turned into a QUBO for asset selection; this QUBO is solved with D-Wave's hybrid solver. Selected assets' weight optimization is done with classical solvers. Using historical Indian market data, they compare against real fund manager performance. Results show the QA-hybrid pipeline can produce diversified portfolios with competitive returns. (Emphasis is on feasibility, not quantum speedup.) [12]. (ArXiv, 2025) link.
- **A real-world test of portfolio optimization with quantum annealing** – *Sakuler et al. (2025)*: Reports a collaboration with a bank, solving their production portfolio optimization problem via QUBO and annealing. They formulate the portfolio (maximizing return under risk constraints) as QUBO, including a novel encoding for continuous weights. Solutions are obtained via two D-Wave hybrid solvers and a purely classical solver. The QA hybrids find solutions consistent with the global optimum, validating QA on a real asset dataset. They also address tuning penalty weights and inequality constraints in QUBO [13]. (Quantum Machine Intelligence) link.

Machine Learning Applications

- **Quantum annealing for feature selection, instance selection, and clustering (QuantumCLEF 2025)** – *Pomeroy et al. (2025)*: Explores three ML tasks via QA (feature and instance selection, clustering). Each task is encoded as a QUBO: for example, feature selection QUBOs balance feature importance vs. redundancy. They run both QA and classical simulated annealing on these QUBOs. Results indicate QA produces competitive solutions efficiently, making it a promising tool for combinatorial ML problems even on current hardware [14]. (CLEF 2025 Notebook) link.
- **Quantum annealing for feature selection in ML** – *Pranjic et al. (2024)*: Studies feature selection by maximizing mutual information (MI) or conditional MI. They construct a *Mutual Information QUBO* (MIQUBO) and solve it on a quantum annealer. Applied to a real-world dataset (predicting used excavator prices), QA-selected feature subsets (especially when MI is sparse) yield significant accuracy improvements over classical MI-based selection. This demonstrates QA's utility in combinatorial feature selection [15] [16]. (ArXiv, 2024) link.
- **Hybrid QA method for generating ensemble classifiers** – *Yulianti et al. (2023)*: Focuses on ensemble learning, where better accuracy is achieved by combining diverse classifiers. They improve the clustering step of ensemble generation by a hybrid QA approach: first form large, strong clusters (by varying clustering parameters), then **optimize pure-class clusters with QA**. Tested on 10 UCI datasets, their method yields more accurate ensembles than baselines. Thus, QA (in a hybrid pipeline) is shown to enhance cluster-based ensemble construction [17]. (J. King Saud Univ. Comput. & Info. Sci.) link.
- **Quantum multiclass SVM (QMSVM) via quantum annealing** – *Delilbašić et al. (2023)*: Proposes a novel QUBO formulation that encodes a multiclass support-vector machine directly (not via one-vs-all ensembles). The entire multiclass classification is formulated as one QUBO solved on D-Wave Advantage. Experiments on remote-sensing data show that QMSVM achieves comparable accuracy to classical SVMs while scaling much better with training set size (almost constant inference time) [18] [19]. This hybrid quantum-classical approach demonstrates feasible end-to-end learning with current annealers. (ArXiv, 2023) link.

NP-hard Graph and Combinatorial Problems

- **Solving the Traveling Salesman Problem on a quantum annealer** – *Jain (2021)*: Implements the standard TSP as a QUBO and solves it on D-Wave's 5000-qubit Advantage machine. The study shows that QA can handle tours of up to ~8 nodes; beyond that, hardware limitations and embedding overhead become prohibitive. In tested instances (5–8 cities), the QA runs found valid tours, but were slower and less optimal than classical solvers [20]. This work provides a thorough QA-based TSP implementation and performance analysis. (Frontiers in Physics) link.
- **Solving the Steiner Traveling Salesman Problem with quantum annealing** – *Ciacco et al. (2025)*: Addresses the *Steiner TSP*, a variant where additional (Steiner) nodes may be visited to reduce cost. The authors reduce the network size via a preprocessing step to fit QA hardware, then encode the problem as a QUBO for a D-Wave annealer. Experiments show the reduction drastically cuts QUBO complexity, enabling the hardware to find high-quality tours. This demonstrates the potential of QA (with careful preprocessing) for complex TSP variants [21]. (ArXiv, 2025) link.
- **Quantum annealing for the selective (orienteering) TSP** – *Le et al. (2021)*: Studies the "selective TSP" where the path must maximize collected score under time budgets. They propose QUBO encodings for selective TSP and solve them on a D-Wave 2000Q. Despite hardware constraints, QA

found optimal tours on several instances by careful problem scaling. This work illustrates encoding and solving an NP-hard routing problem in QA form [22]. (Proc. SPIE 2021) link.

- **Accuracy and performance of QA on Max-Cut** – *Vodeb et al. (2024)*: Benchmarks D-Wave's latest QA hardware and hybrid solvers on Max-Cut instances (100–10,000 nodes). The study compares solution quality and time to classical simulated annealing and Toshiba's simulated bifurcation machine. Results show that for small graphs (<250 nodes), the hybrid QPU and SA find global optima more reliably than QPU alone, while for larger graphs non-QPU methods (bifurcation, slow SA) are competitive. This paper provides up-to-date empirical insight into QA vs classical on a canonical NP-hard problem [23]. (ArXiv, 2024) link.
- **Graph partitioning using a quantum annealer** – *Lucas et al. (2017)*: Early work on graph partitioning (balancing vertex sets) via D-Wave. The modularity matrix (or adjacency) is thresholded to reduce qubit count, and recursive strategies are used for k-way partitioning. Experiments on various graphs (including sparse Walshaw benchmarks) found that QA (and QA+hybrid qbsolv) matches or outperforms classical tools (METIS, KaHIP) on cut size in many cases [24]. This demonstrates that formulating graph partition as a QUBO can yield competitive results: "GP framed as a QUBO is a natural fit for the D-Wave annealer" [24]. (Int'l Conf. on Quantum Annealing 2017; ArXiv) link.

Sources: Key papers as cited above. Each entry cites the corresponding source via the bracketed links (e.g., [1]).

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